Sheet 22 Thu. 10 Dec. 2026 Square root and cube; sign of the reference functions

1. Warm-up · Flash Calculate $\overline{49}$, $\sqrt{0.25}$ and $(-2)^3$.

2. Core · Method Study the sign of $f(x) = -2x + 6$.

3. Core + · True/false True or false? Justify. For every real number x, $\sqrt{x^2} = x$.

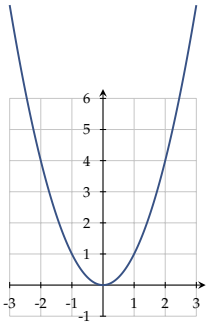
4. Back · Flash How many squares make up a net of a cube? A cross (row of 4 squares + 1 squaring both sides) work?

5. Running thread · Flash How many faces, vertices and edges does a square-based pyramid have?

Sheet 23 Mon. 14 Dec. 2026 Graphical and numerical solving: f $(x) = k$, $f(x) < k$

1. Warm-up · Flash Solve $x^2 = 16$ then $x^2 = -4$.

2. Core · Graph Curve of $f(x) = x^2$:



Solve graphically $f(x) = 4$ then $f(x) < 4$.

3. Core + · Method Solve $|x| < 3$ then $|x| = 5$.

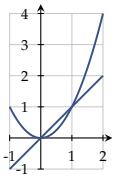
4. Back · Trap Calculate $-x^2$ then $(-x)^2$ for $x = -3$.

5. Running thread · Method In a triangle ABC, M and N are the midpoints of $[AB]$ and $[AC]$. What is being proved? Which property is being cited?

Sheet 24 Tue. 15 Dec. 2026 Solving f $(x) = g(x)$ *et* $f(x) < g(x)$

1. Warm-up · Flash Solve 2 $x - 1 = x + 3$.

2. Core · Graph Graphs of $f(x) = x$ and $g(x) = x^2$:



Solve $f(x) = g(x)$ then $f(x) < g(x)$.

3. Core + · Reading code What does `f(-1) == g(-1)` ? And `f(0) == g(0)` ?

```
def f(x):
    return x*x
def g(x):
    return 2*x + 3
```

4. Back · Problem Plan A: 15,000 Ar fixed + 1,000 Ar per session. Plan B: 2,500 Ar per session. At from how many sessions is A cheaper?

5. Running thread · Flash Calculate the coordinates of the midpoint of $[AB]$ with $A(-2;5)$ and $B(6;1)$.

One-variable descriptive statistics

Sheet 25 Thu. 17 Dec. 2026 Mean and linearity of the mean

1. Warm-up · Flash Mean of 12, 15, 9, 14?

2. Core · Method The data set 4, 6, 8, 10, 12 has mean 8. What is the mean if we add 3 to each value? if each value is multiplied by 2?

3. Core + · True/false True or false? Justify. The mean of two numbers $a < b$ is always strictly between a and b.

4. Back · Reading In a pie chart of 80 people, one sector represents 25%. What is its frequency? What is its angle?

5. Running thread · Flash Calculate gcd $(84\,;\,126)$ and deduce the irreducible form of $\dfrac{84}{126}$.

Sheet 26 Mon. 4 Jan. 2027 Median and quartiles

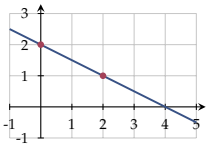
1. Warm-up · Flash Order 7, 2, 9, 4, 5, 11, 3 then give the median.

2. Core · Method Series 3 ; 5 ; 6 ; 8 ; 9 ; 12 ; 14 ; 15. Give Q_1, the median, Q_3 and the interquartile range.

3. Core + · True/false True or false? Justify. At least half of the values of a series are greater or equal to its median.

4. Back · Flash Order 0.6, $\dfrac{5}{8}$ and 61 % in increasing order.

5. Running thread · Graph → formula The line below represents an affine function f. Determine $f(x)$.



Sheet 27 Tue. 5 Jan. 2027 Standard deviation; effect of adding or removing a value

1. Warm-up · Flash Calculate $(3-5)^2+(7-5)^2$.

2. Core · Method Series 2 ; 4 ; 6 : mean, variance and standard deviation (approximate value to 10^{-2}).

3. Core + · True/false True or false? Justify. Adding 5 to each value of a series does not change its standard deviation.

4. Back · Construction steps Let ABC a triangle, I the midpoint of $[AB]$ and D the reflection of C by relative to I. What type of $ACBD$?

5. Running thread · Flash $(MN) \parallel (BC)$, $M \in [AB]$, $N \in [AC]$, $AM = 3$, $AB = 5$, $AC = 10$. Calculate AN.

Sheet 28 Thu. 7 Jan. 2027 Comparison of series

1. Warm-up · Flash Range of 3, 8, 5, 12?

2. Core · Interpret Two series A and B have the same mean 12; $\sigma_A = 1.5$ and $\sigma_B = 4$. Which is the more regular?

3. Core + · Interpret A : median 11, interquartile range 2. B : median 13, interquartile range 6. Compare the two series.

4. Back · Flash Two fair coins are tossed. Probability of getting two tails?

5. Running thread · Method Procedure: choose a number; square it; add 1. Result for -3? Can we get 0? Justify.